VHDL Condition

A condition is an expression that evaluates to a Boolean result, true or false.

Example:

SIGNAL s1, s2, s3 : STD_LOGIC;

IF (s1 = '1' AND s2 = '0 AND s3 = '1') THEN . . .

The condition s1 = '1' AND s2 = '0 AND s3 = '1' evaluates to true if s1 is '1' and s2 is '0' and s3 is '1'.

VHDL 2008 condition enhancement

VHDL 2008 includes the c*ondition operator* "??" that converts a BIT or STD_LOGIC value to a BOOLEAN value; BIT value '1' is converted to TRUE and '0' to FALSE; STD_LOGIC value '1' and 'H" are converted to TRUE, and all other values to FALSE.

Example:

IF ((?? s1) AND (?? (NOT s2)) AND (?? s3)) THEN . . .

which is logically equivalent to IF (s1 = '1' AND s2 = '0 AND s3 = '1') THEN . . .

Implicit use of ?? operator

If the "??" operator is used implicitly, it must be applied to the entire condition and the entire condition must contain only conditions of the same type.

Example:

IF (s1 AND (NOT s2) AND s3) THEN . . . supported by Quartus version 11

The following implicit use of ?? operator is illegal:

IF (s1 AND (NOT s2) AND s3 = '1') THEN . . . NOT OK because of s1 combined with s3 = '1'

IF (s1 AND (NOT s2) AND s3 ?= '1') THEN . . . OK but still not supported by Quartus 11.0

The implicit use of ?? operator can be applied in these constructs: IF, IF THEN ELSE, WHEN ELSE, WAIT UNTIL, ASSERT, WHILE, etc.

Example:

WAIT UNTIL (NOT s1 OR s2); equivalent to WAIT UNTIL (s1 = '0' OR s2 = '1');